KPR INSTITUTE OF ENGINEERING AND TECHNOLOGY

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

EXPLORATORY DATA ANALYSIS AND DEEP LEARNING-BASED

PREDICTION OF HOTEL BOOKING CANCELLATIONS

Assignment II

Student Name: DURGESH G V

Roll Number: 24AD033

Year / Semester: III / 05

Course Code: U21ADP05

Course: Exploratory Data Analysis and Visualization

Submission Date: 15 October 2026

Abstract

This project explores the Hotel Booking Demand dataset and develops a Deep Learning model to predict whether a hotel booking will be cancelled. The analysis uses Python, Pandas, Seaborn, Matplotlib and TensorFlow/Keras. Data preparation includes removing duplicate records, handling selected missing values, creating total-night and total-guest features, and encoding categorical variables while scaling numerical variables. Descriptive visualizations examine cancellation balance, booking lead time, average daily rate, hotel type, numerical correlations and deposit type. A multilayer perceptron (MLP) is trained using training and validation data, with a separate test set used for final evaluation. The model achieved 84.36% accuracy and a ROC-AUC of 0.909 on the test set. The findings suggest that booking lead time and deposit type are useful signals, although relationships in historical data do not prove causation.

1. Introduction and Objectives

Hotel booking cancellations can affect room planning, revenue and staff scheduling. Exploring booking records can help identify patterns associated with cancellation. The objective of this project is to perform exploratory data analysis (EDA), visualize important patterns, preprocess the data, and train a Deep Learning model to predict the cancellation status.

Objectives:

- Inspect the dataset and summarize its important fields.
- Clean duplicate records and handle selected missing values.
- Create at least five visualizations and interpret the patterns.
- Train and evaluate an MLP classification model.
- Discuss limitations and possible future improvements.

2. Dataset Description

The dataset is Hotel Booking Demand, available from Kaggle: https://www.kaggle.com/datasets/jessemostipak/hotel-booking-demand. Published descriptions report 119,390 records and 32 original columns. The notebook checks the uploaded CSV's actual dimensions. Important fields include hotel, lead_time, arrival-date information, guest counts, deposit_type, adr, total_of_special_requests and is_canceled. The target variable is is_canceled: 0 means not cancelled and 1 means cancelled.

3. EDA and Preprocessing

The data was loaded using Pandas. The notebook inspected column types, summary statistics, missing values and duplicate rows. Exact duplicate rows were removed. Missing children values were filled with zero, while missing country, agent and company values were labelled as Unknown. Rows with missing or negative ADR were excluded. Two additional features were created: total_nights and total_guests. The interquartile range (IQR) rule was used to flag possible outliers in lead_time, adr and total_nights. These flags were inspected rather than automatically deleting all extreme values, because unusual bookings may still be valid.

The target column is is_canceled. To prevent target leakage, reservation_status and reservation_status_date were excluded from the predictors. The data was divided into training (70%), validation (15%) and test (15%) sets using stratified splitting. Numerical missing values were imputed with the median and scaled using StandardScaler. Categorical missing values were imputed with the most frequent value and encoded using one-hot encoding. The preprocessing pipeline was fitted on training data only.

4. Seven Stages of Visualization

The assignment PDF asks for all seven stages of visualization but does not list their names. This project documents the following practical workflow: (1)

define the question, (2) collect data, (3) prepare data, (4) choose chart types and encodings, (5) create charts, (6) interpret findings, and (7) communicate results. If the course instructor has specified a particular seven-stage framework, its exact terminology should be used in the final submission.

5. Data Visualizations and Interpretations

5.1 Booking Cancellation Distribution



Figure 1. Distribution of booking cancellation status.

What it shows: the number of bookings in each target class. Why created: to understand the class balance before modeling. Insight: the chart shows more bookings were not cancelled than cancelled, so the target classes are imbalanced.

5.2 Booking Lead-Time Distribution

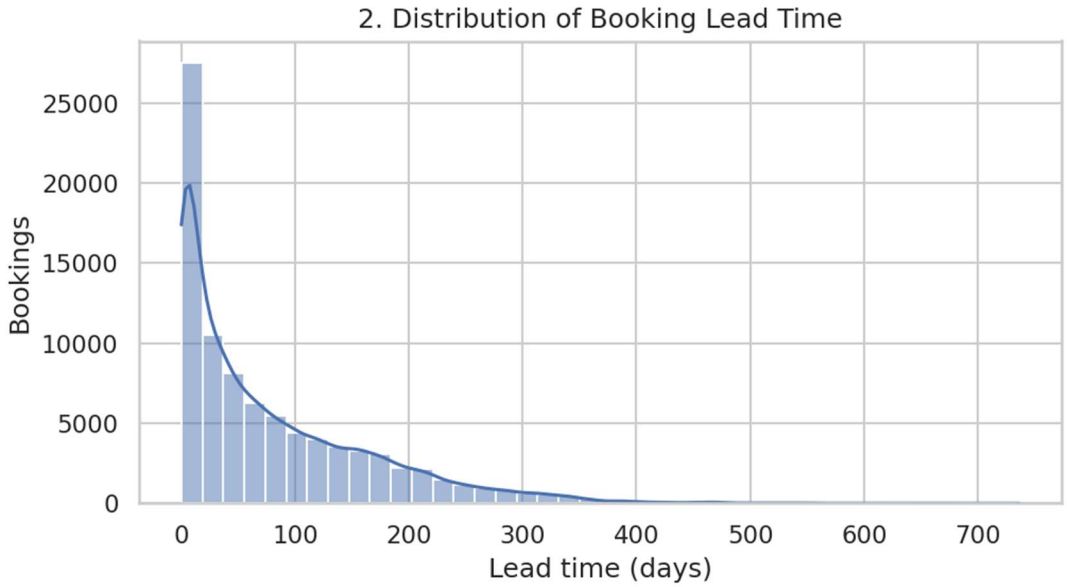


Figure 2. Distribution of lead time.

What it shows: how many bookings were made at different numbers of days before arrival. Why created: to inspect the distribution of a numerical feature. Insight: the distribution is strongly right-skewed; many bookings have shorter lead times, while fewer bookings have very long lead times.

5.3 Average Daily Rate by Cancellation

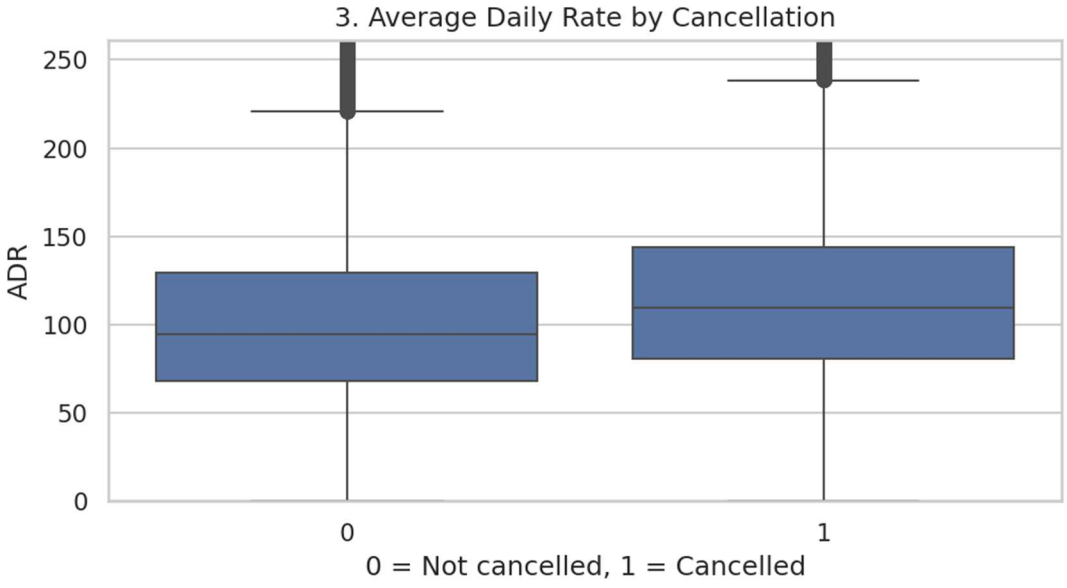


Figure 3. Average daily rate (ADR) by cancellation status.

What it shows: the median, spread and possible outliers of ADR for cancelled and non-cancelled bookings. Why created: to compare a numerical feature across the two classes. Insight: the cancelled group appears to have a higher median ADR in this plot, and both groups contain high-value observations.

5.4 Cancellation Rate by Hotel Type

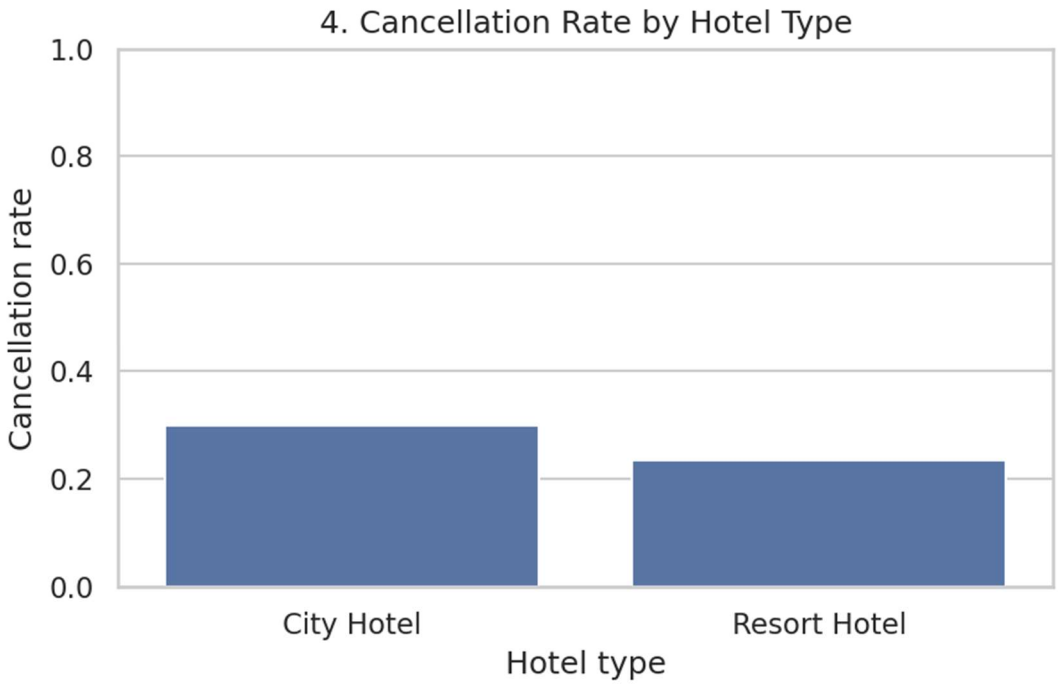


Figure 4. Cancellation rate for City Hotel and Resort Hotel.

What it shows: the proportion of bookings cancelled for each hotel type. Why created: to compare cancellation behaviour across categories. Insight: the City Hotel cancellation rate is higher than the Resort Hotel rate in this dataset. This is an association, not proof that hotel type causes cancellation.

5.5 Correlation Heatmap

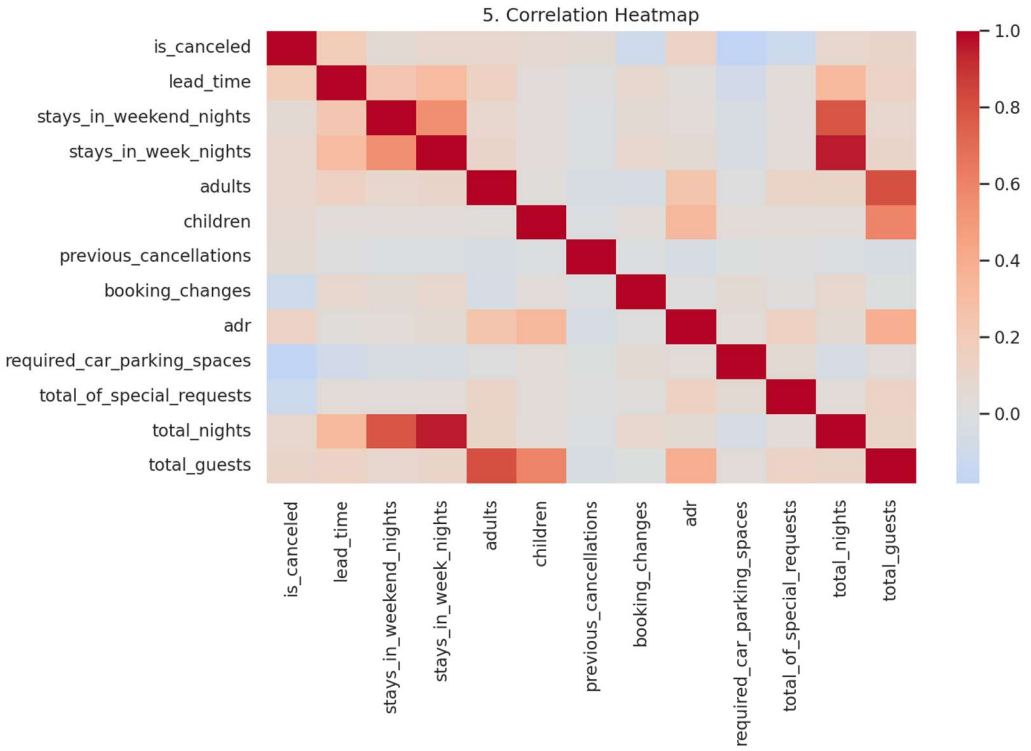


Figure 5. Correlation heatmap of numerical features.

What it shows: pairwise correlations between selected numerical variables. Why created: to explore relationships and possible redundant features. Insight: total_nights is strongly related to the component stay-night fields, and total_guests is related to guest-count fields because these engineered variables are sums of their components.

5.6 Lead Time versus ADR

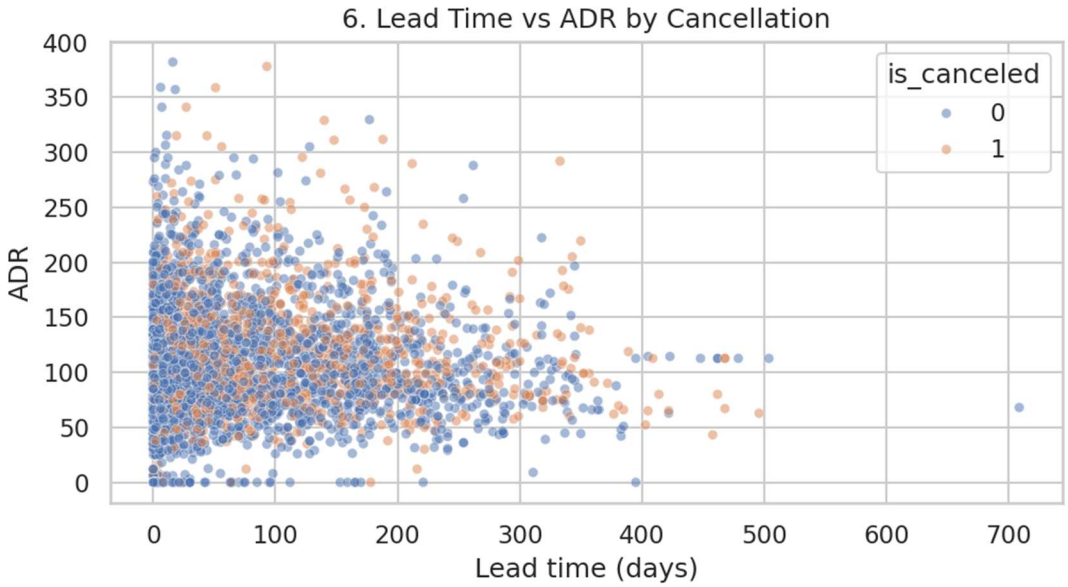


Figure 6. Lead time versus ADR, coloured by cancellation status.

What it shows: the relationship between booking lead time and ADR for a sample of bookings. Why created: to explore a possible bivariate relationship and compare classes. Insight: observations overlap substantially, so these two variables alone do not clearly separate all cancelled and non-cancelled bookings.

5.7 Cancellation Rate by Deposit Type

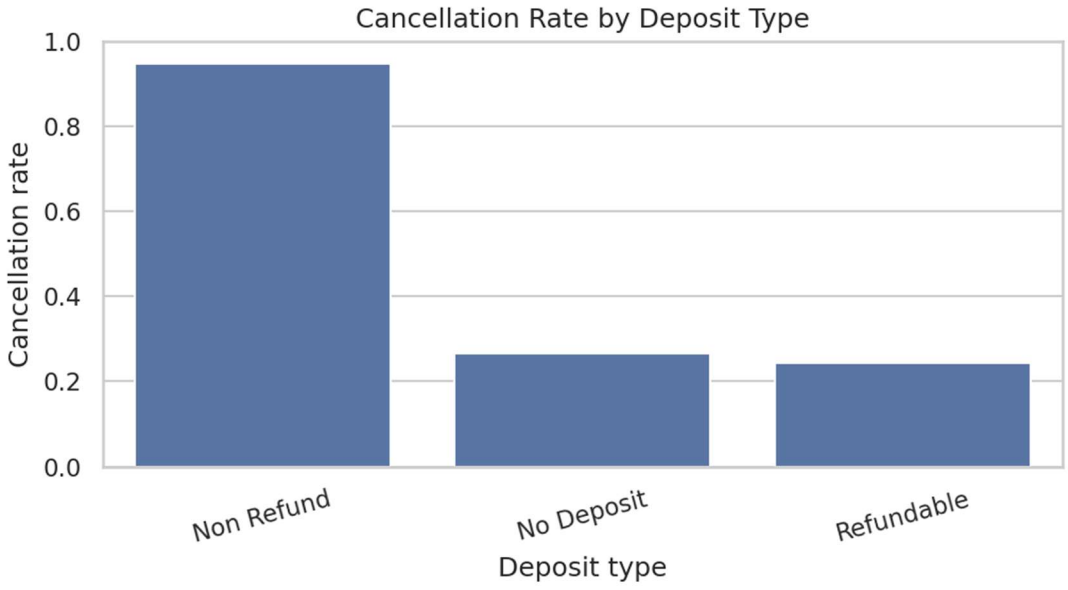


Figure 7. Cancellation rate by deposit type.

What it shows: cancellation proportions for each deposit category. Why created: to examine whether booking deposit category is associated with cancellation. Insight: the Non Refund deposit category has a much higher observed cancellation rate in this dataset. The category may reflect specific booking conditions and should not be interpreted as a causal effect.

6. Deep Learning Model

A multilayer perceptron (MLP) was implemented using TensorFlow/Keras. The model accepts the preprocessed feature vector, uses a Dense layer with ReLU activation, a Dropout layer, another Dense layer with ReLU activation, and a final Dense layer with one sigmoid output. The sigmoid output represents the estimated probability that a booking is cancelled.

The model used the Adam optimizer and binary cross-entropy loss. Training was limited to 15 epochs with a batch size of 256, and early stopping monitored validation loss. Two configurations were compared using validation ROC-AUC: 64 or 128 units in the first hidden layer, with different dropout rates. The configuration with the higher validation ROC-AUC was selected.

7. Model Evaluation and Results

The selected model was evaluated on the held-out test set. The observed metrics were:

Metric	**Test result**
Accuracy	84.36%
Precision	73.49%
Recall	67.40%
F1-score	70.31%
ROC-AUC	0.9087

Accuracy of 84.36% means that about 84 out of every 100 test predictions were correct. Precision of 73.49% means that, among bookings predicted

as cancelled, about 73% were actually cancelled. Recall of 67.40% means that the model identified about 67% of the bookings that were actually cancelled. The F1-score of 0.703 balances precision and recall. ROC-AUC of 0.9087 indicates good ability to rank cancelled bookings above non-cancelled bookings across thresholds, although it does not mean that 90.87% of predictions are correct.

7.1 Loss versus Epoch

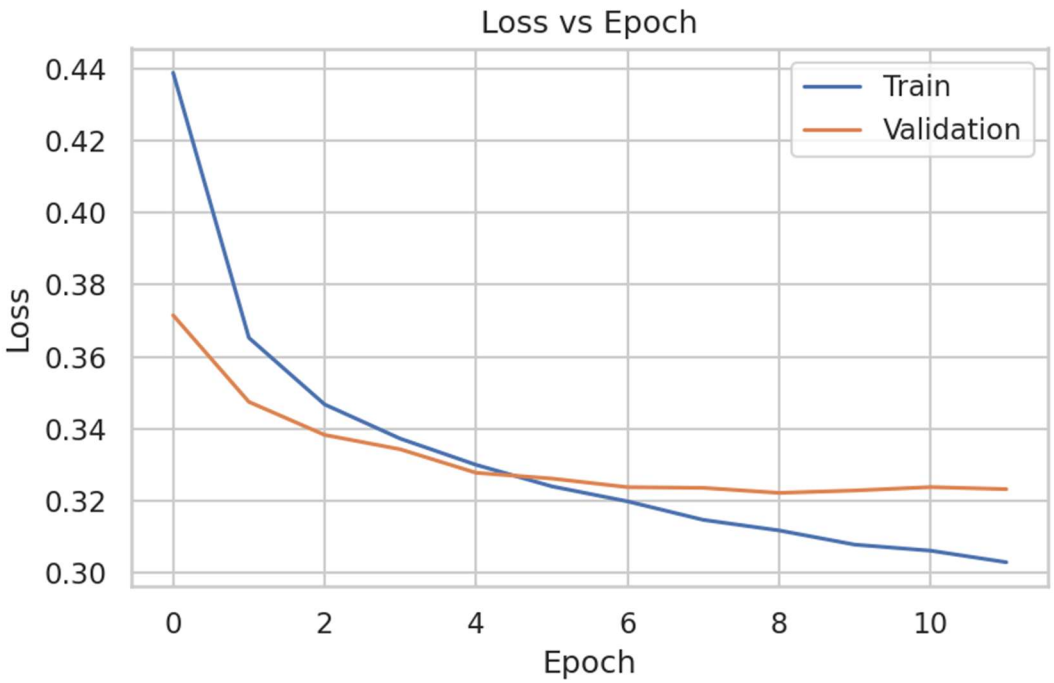


Figure 8. Training and validation loss by epoch.

The loss chart shows how prediction error changes during training. A decreasing training loss indicates learning; the validation curve helps identify whether generalization stops improving.

7.2 Accuracy versus Epoch

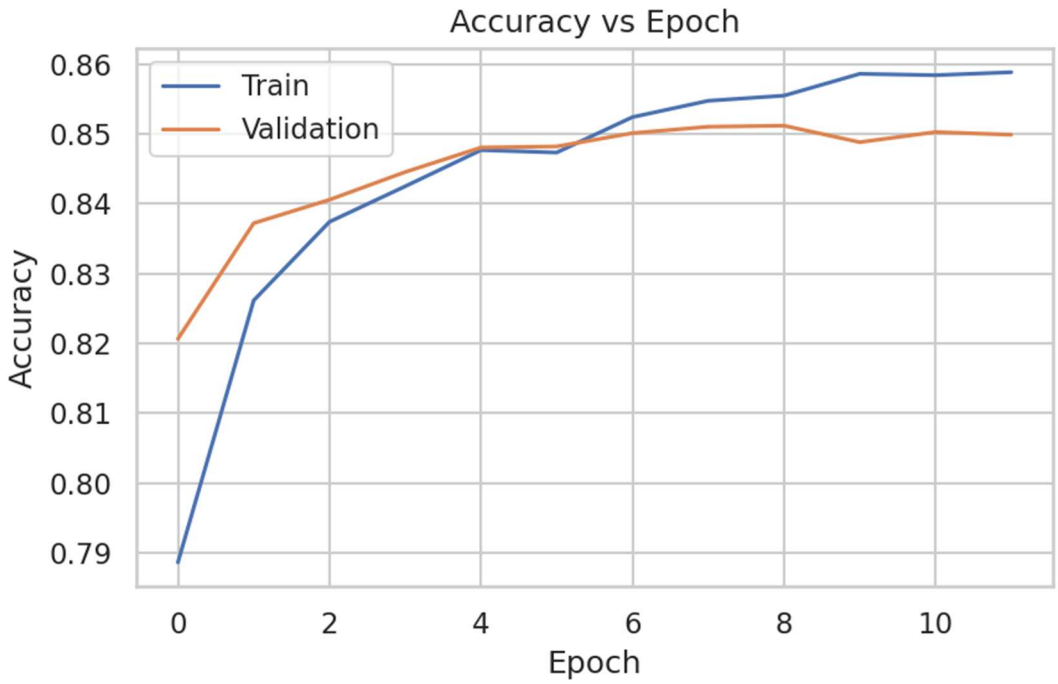


Figure 9. Training and validation accuracy by epoch.

The accuracy chart compares performance on training and validation data over epochs. Large persistent gaps between the curves can indicate overfitting.

7.3 Confusion Matrix

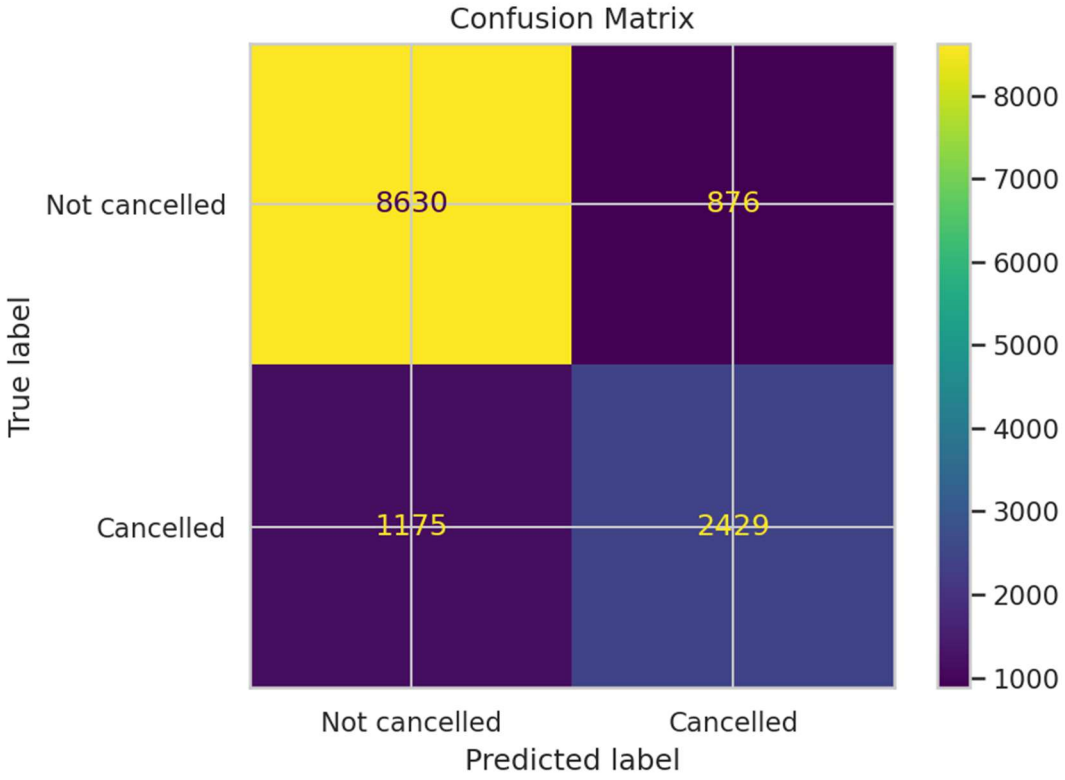


Figure 10. Confusion matrix on test data.

The confusion matrix reports 8,630 correctly identified non-cancelled bookings, 876 non-cancelled bookings incorrectly predicted as cancelled, 1,175 cancelled bookings incorrectly predicted as not cancelled, and 2,429 correctly identified cancellations. The model therefore misses some real cancellations, which is reflected in its recall.

7.4 ROC Curve

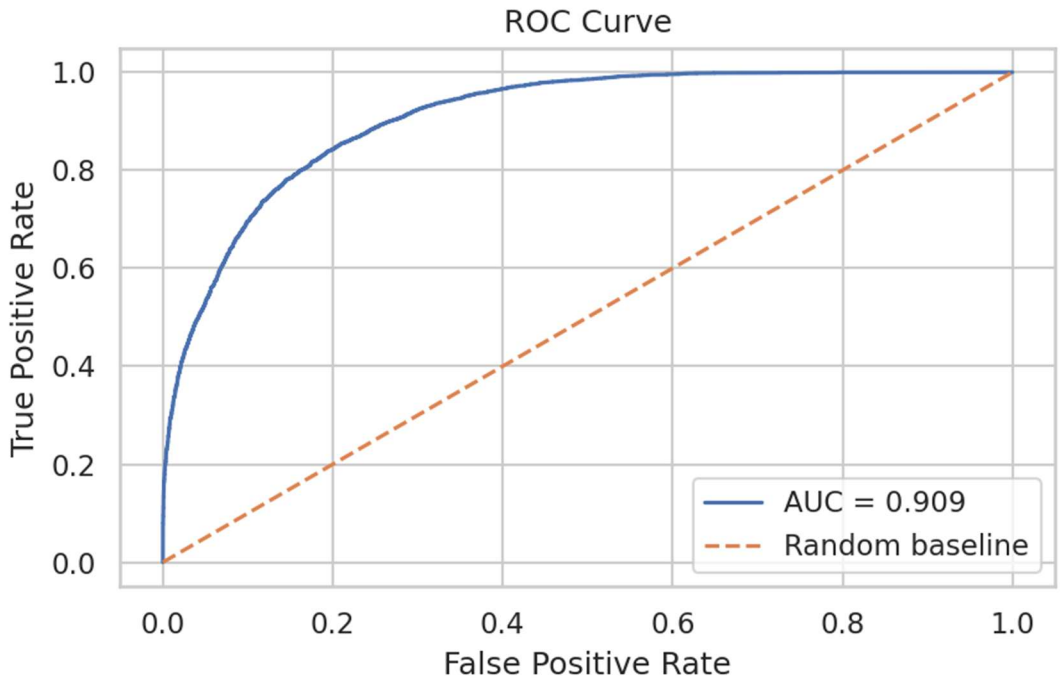


Figure 11. ROC curve on the test set.

The curve plots true-positive rate against false-positive rate at different thresholds. The test ROC-AUC is 0.9087, which indicates good discrimination between the two classes.

7.5 Approximate Feature Importance

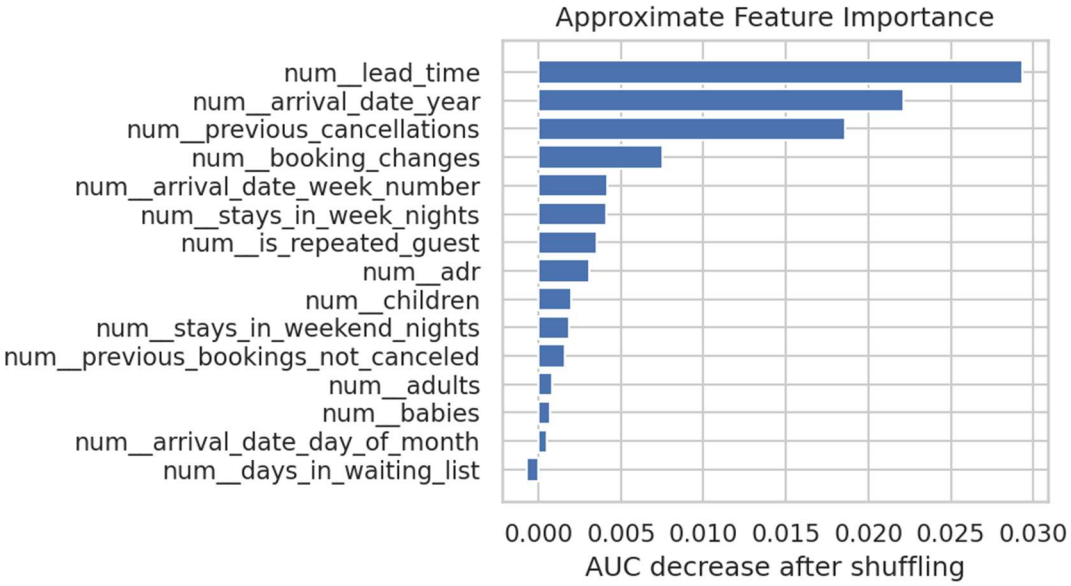


Figure 12. Approximate permutation feature importance.

Features are shuffled one at a time, and the decrease in ROC-AUC is measured. Lead time, arrival-date year and previous cancellations appear among the more influential encoded features in this run. This is an approximate importance measure, not evidence of causation.

8. Conclusion and Future Scope

This project applied EDA, statistical summaries, visualizations and Deep Learning to hotel booking data. The charts showed that cancellation rates differ by hotel type and deposit type, the lead-time distribution is right-skewed, and some numerical features are strongly related because they are derived from the same underlying fields. The MLP achieved 84.36% test accuracy and a ROC-AUC of 0.9087. Its recall was 67.40%, meaning some actual cancellations were missed. Therefore, the model may help identify cancellation risk but should not be used as the only basis for hotel decisions.

Limitations include reliance on historical data, possible changes in customer behaviour and hotel policies, class imbalance, and the fact that correlations do not establish cause and effect. Future work could compare the MLP with tree-based models, tune the decision threshold according to business costs, test on newer data, and monitor model performance over time.

9. References

1. Antonio, N., de Almeida, A., & Nunes, L. (2019). Hotel booking demand datasets. Data in Brief, 22, 41–49. https://doi.org/10.1016/j.dib.2018.11.126

2. Kaggle. (n.d.). Hotel booking demand [Data set]. https://www.kaggle.com/datasets/jessemostipak/hotel-booking-demand

3. TensorFlow. (n.d.). The Sequential model. https://www.tensorflow.org/guide/keras/sequential_mode

4. Keras. (n.d.). EarlyStopping. https://keras.io/api/callbacks/early_stopping/
.

10. Appendix — Source Code

The complete Python source code is maintained in the project's GitHub repository. The repository should include the source code notebook/file, project report, README, dataset or dataset source link, and generated output figures. Add the public GitHub repository URL here:

https://github.com/Durgesh-GV/Prediction-of-Hotel-Booking-Cancellations.git